

Trading Strategies, Portfolio Monitoring and Rebalancing¹

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ABSTRACT

Trading strategies translate goals and constraints of asset management into dynamic, intertemporal, coherent portfolio decisions. Under special assumptions, myopic portfolio policies are shown to be optimal and constant over time. In general, however, both optimal portfolio and current positions are subject to random movements so that periodic monitoring and rebalancing are called for. Transaction costs and monitoring costs create a trade-off between the costs of tracking error with respect to the optimal allocation and the costs to implement it into current portfolio. The results of the theory of optimal rebalancing show that the optimal allocation is replaced by a no-trade region delimited by rebalance boundaries. The factors influencing the boundaries and the rebalance decisions can be analytically and numerically explained. Popular rebalancing rules imply a substantial amount of excess trading costs but they could generate positive net returns in the case of mean-reverting market regimes.

KEYWORDS: Trading strategies; transaction costs; rebalancing; benchmarking

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INTRODUCTION

Trading strategies represent the core and the engine of portfolio management.

Both active and passive management imply adequate trading strategies, with the only exception of the “buy and hold” strategy (BH), which is simply defined as the implementation and maintenance of the initial quantity of the assets purchased at time 0.

All trading strategies are defined with respect to goals and constraints and imply a continuous monitoring activity in order to assess the effectiveness of the strategy, its coherence and compliance, its distance from the stated objective of the asset management.

Rebalancing is the typical result of portfolio monitoring and it consists of a set of trading orders aimed at repositioning the portfolio for a better accomplishment of both goals and constraints.

There are, in fact, a number of different reasons by which rational investors want to trade in financial markets and adjust their financial portfolios: one is market movements, to be distinguished in noisy (transitory) and structural (persistent); a second one is modification of individual characteristics, among which personal wealth is certainly of the greatest importance, a third one is new information, expectations and learning, suggesting improved portfolios, a fourth one is just the passing of time, making previous portfolio positions no longer optimal (as in portfolio insurance or life-cycle investments).

[Fig. 18.1 about here]

Suppose that an optimal (percentage) asset allocation $w^*(t)$ has been worked out. It is then possible to distinguish between different portfolio adjustments. If $w_t(t+1)$ is the portfolio allocation obtained with time t quantities and time $t+1$ prices (see the Appendix) the allocation dynamics from t to $t+1$ are the following:

$$w_{t+1}(t+1) = w_t(t) + [w_t(t+1) - w_t(t)] + [w^*(t) - w_t(t+1)] + [w^*(t+1) - w^*(t)] + [w_{t+1}(t+1) - w^*(t+1)]$$

where the four components in brackets represent, respectively, the change from price movements (**passive rebalancing**), the distance from the old target $w^*(t)$, the change in the optimal target

(**target rebalancing**), the distance between the new portfolio and the new target $w^*(t+1)$ (**portfolio rebalancing**).

As documented in Calvet, Campbell and Sodini (2007), using a large panel of Sweden households, passive rebalancing in the period 1999-2002 has been offset for about one half by the other kind of rebalancing (active rebalancing), with higher reactions in the case of more affluent investors in terms of wealth, education and income. Cremers and Mei (2004) confirm the risk control motivation behind about three quarter of total asset trading.

Simplifying the first components we have:

$$w_{t+1}(t+1) = w^*(t) + [w^*(t+1) - w^*(t)] + [w_{t+1}(t+1) - w^*(t+1)]$$

and even in the case of constant target w^* , portfolio rebalancing is required to offset the price movements.

In current practice, this is in fact the usual meaning of “rebalancing”, that is the trading activity by which an optimal (strategic) allocation among different asset classes (target constant mix, CM), altered by the market movements of relative prices, is restored by selling the growing assets and buying the dropping securities (“sell high and buy low”).

Two questions arise concerning this meaning of rebalancing:

- 1) is restoration of the target mix a good idea? i.e. is the original asset allocation still optimal? or, are market relative movements void of useful information and just an effect of noise and randomness? A negative answer to these questions would imply passing from static to dynamic asset allocation, including target rebalancing as an endogenous choice into the general allocation process with adjustment costs;
- 2) what are the terms of the trade-off between transaction costs and tracking errors, making rebalancing a proper optimization problem?

Clearly, in the ideal situation of frictionless markets, continuous rebalancing would be costless and it would guarantee a zero tracking error with respect to any target mix. With transaction costs, rebalancing should make a balance between two negative effects: the distance from target and the cost of reducing it.

In the following, we shall review the answers to these and other related questions provided in many decades of financial literature.

MYOPIC PORTFOLIOS AND THE CASE FOR REBALANCING

Trading strategies (TS), both static and dynamic, can be classified, with respect to their objectives, into two broad classes: absolute and relative trading strategies.

Absolute TS have an absolute goal which can be expressed in terms of risk or in terms of returns. Absolute TS with a target risk are TS whose objective is to reach and maintain a given level of risk, suitably defined. They maximize the portfolio return under the constraint of (maximum) risk. Absolute TS with a target return are TS whose objective is to reach and maintain a given level of (total) return. They minimize the portfolio risk under the constraint of a given (minimum) return. Relative TS have a relative goal which is often expressed in terms of a benchmark portfolio.

[Fig. 18.2 about here]

The seminal reference on static, absolute trading strategies is Markowitz's (1952, 1959) Nobel prize result of the mean-variance portfolio selection. According to his static analysis, considering one single-period (0 to T) horizon and just two assets for simplicity, one riskless and one risky, the optimal risky position is given by:

$$w_{0,T} = \frac{\mu_{0,T} - r_{0,T}}{\vartheta \sigma_{0,T}^2} \quad (18.1)$$

where ϑ is the investor's risk aversion parameter and $\mu_{0,T}$ ($r_{0,T}$) is the expected value of the risky (riskless) rate of return over the 0-T horizon and $\sigma_{0,T}^2$ is its variance.

If the 0 to T horizon is composed by T unit periods ("months") in which the investor has full access to the financial markets to buy and sell securities (but see Longstaff, 2009 for major effects of nonmarketability and blackout periods) the same analysis would produce for the first period:

$$w_{0,1} = \frac{\mu_{0,1} - r_{0,1}}{\vartheta \sigma_{0,1}^2} \quad (18.2)$$

and, under the heroic assumption of (serially) independent, identically distributed $D(\mu, \sigma^2)$ rates of returns (IID hypothesis), the same allocation (Markowitz portfolio) would be optimal for every single and the overall period:

$$w^* = \frac{\mu - r}{\vartheta \sigma^2} = \frac{T\mu - Tr}{\vartheta T\sigma^2} = \frac{\mu_{0,T} - r_{0,T}}{\vartheta \sigma_{0,T}^2} \quad (18.3)$$

This means that no target rebalancing is required from one period to the next and the initial one-period allocation is optimal for good. In an intertemporal setting with a long run horizon, this is clearly a myopic policy: look at the near even if the target is further.

If returns are not IID this is in general a sub-optimal dynamic policy and there is room for optimally rebalancing the allocation from one period to the next toward the global time horizon.

One famous exception, pointed out by Mossin (1968) and Fama (1970), is represented by the investor with log-utility in final wealth. As shown in the Appendix, the portfolio value at time T can be expressed as:

$$W(T) = W(0) \prod_{t=0}^{T-1} \sum_{j=1}^N w_{j,t} (1 + R_j(t+1)) \quad (18.4)$$

so that log-utility is the sum of T logs which can be maximized separately (i.e. myopically):

$$\begin{aligned} \max_{w_0, w_1, \dots} \{E_o(\log(W(T)))\} &\Rightarrow \max_{w_0, w_1, \dots} \left\{ \sum_{t=0}^{T-1} E_o \left(\log \left(\sum_{j=1}^N w_{j,t} (1 + R_j(t+1)) \right) \right) \right\} \\ &\Rightarrow \sum_{t=0}^{T-1} \max_{w_t} \left\{ E_t \left(\log \left(\sum_{j=1}^N w_{j,t} (1 + R_j(t+1)) \right) \right) \right\} \end{aligned} \quad (18.5)$$

This means that in each period t the investor has to maximize $E_t \log (W(t+1) / W(t))$ i.e the expected continuously compounded rate of growth of wealth ("growth-optimal policy"). Rebalancing should take place from one period to the next but, under the additional hypothesis of log-normal distribution for values, we have:

$$E_t \log \left(\frac{W(t+1)}{W(t)} \right) = r + w(\mu - r) - \frac{1}{2} w^2 \sigma^2 \quad (18.6)$$

and the optimal myopic policy is again the constant Markowitz portfolio w^* with $\vartheta=1$.

Merton (1969, 1971) has generalized this result under the now classical setting of continuous time, continuous trading and diffusion processes. With only two assets, assuming that asset prices are governed by the following general dynamics:

$$\begin{aligned}\frac{dP_1(t)}{P_1(t)} &= \mu(t)dt + \sigma(t)dZ(t) \\ \frac{dP_0(t)}{P_0(t)} &= r(t)dt\end{aligned}\quad (18.7)$$

and the investor maximizes the expected lifetime discounted utility of consumption and final wealth $W(T)$, he showed that the optimal portfolio is given by:

$$w^*(t) = \frac{J_W}{-J_{WW}W(t)} \frac{\mu(t) - r(t)}{\sigma^2(t)} + \frac{J_{1W}P_1(t)}{-J_{WW}W(t)} \quad (18.8)$$

where $J(W, P_1, t)$ is the maximized indirect utility function (Bellman's value function) and J_W, J_{WW}, J_{1W} are partial derivatives. Under price log-normality, J is no longer function of prices, the hedging demand for price movements is zero and we obtain:

$$w^*(t) = \frac{J_W}{-J_{WW}W(t)} \frac{\mu - r}{\sigma^2} \quad (18.9)$$

where time dependence of w^* comes only from the (indirect) relative risk aversion coefficient, that is $-J_{WW}(W, t) W(t) / J_W(W, t)$.

In the general class of discounted hyperbolic absolute risk aversion utilities (HARA):

$$U(W, t) = e^{-\rho t} \frac{1 - \gamma}{\gamma} \left(\frac{\beta W}{1 - \gamma} + \eta \right)^\gamma \quad (18.10)$$

the dynamics of the optimally invested wealth are given by:

$$dW(t) = \left(r + w^*(t)(\mu - r) - \frac{c^*(t)}{W(t)} \right) W(t)dt + \sigma w^*(t) W(t)dZ(t) \quad (18.11)$$

and the optimal portfolio is:

$$w^*(t) = \frac{\mu - r}{(1 - \gamma)\sigma^2} \left(1 + \frac{(1 - \gamma)\eta}{rW(t)\beta} (1 - e^{-r(T-t)}) \right) \quad (18.12)$$

so that continuous rebalancing is required. Possible time paths of the portfolio allocation and its time changes (target rebalances) are given in Fig. 18.3, for suitable values of parameters (in particular: $\mu=5\%$, $\sigma=30\%$, $\gamma=-4$, $r=1\%$) and monthly time steps.

[Fig.18.3 about here]

With constant relative risk aversion (CRRA) utility, $\eta=0$, the optimal intertemporal allocation is again the Markowitz constant portfolio and portfolio rebalancing is required only to offset price movements.

As argued by Campbell and Viceira (2002), CRRA is, in fact, a plausible assumption given that, in a long-run perspective, per capita wealth has greatly increased in many countries during the last century but risk premia have not shown any long term trend, implying that RRA as percentage compensation for bearing the risk (Pratt, 1964) is, *prima facie*, independent of wealth and other trended state variables.

PORTFOLIO ALLOCATION AND REBALANCING WITH TRANSACTION COSTS

In the last, applied chapter of his seminal book, Markowitz (1959) recognizes the issue of illiquidities and transaction costs, arguing that “The Rational Investor must move more slowly toward his desired portfolio because of the existence of these illiquidities” (ivi, p. 300) and suggesting this critical area for important future development: “My own chief theoretical worry remains with the gap between a theory based on perfect liquidity and the existence of illiquidities” (ivi, p. xii).

Bid-ask spread, brokerage fees, trading taxes (e.g. Tobin’s tax and capital gain tax), price impact of sales/purchases and all other kind of transaction costs represent the main sources of illiquidity (Demsetz, 1968, Collins and Fabozzi, 1991).

It may be useful, here, to remind the classical distinction between liquidity and marketability. As pointed out by Hicks (1962), following Keynes (1930), the characteristic of liquid asset is that it can be “certainly realisable at short notice without loss”. The first part of this

definition (“certainly realisable at short notice”) is peculiar to the wider class of marketable assets; the second part (“certainly without loss”) identifies the subset of liquid assets. A liquid asset is also marketable but not viceversa. The presence of transaction costs generates illiquidities even if it is compatible with perfect marketability. See Longstaff (2009) for recent developments and references on this issue.

Early contributions on portfolio selection and transaction costs were Smith (1968), comparing, in a simulation experiment, BH, CM, and two kind of dynamic adjustment and Leland (1974) and Goldsmith (1976), considering fixed transaction costs in a single period model.

However, on the wake of the dynamic approach of Merton’s (1969, 1971, 1973) intertemporal asset allocation model, Magill and Constantinides (1976) represent a path-breaking paper and a cornerstone result on the subject.

A part from a technical error pointed out later by Duffie (1983) and corrected in Constantinides (1986), they provided for the first time a general framework to endogenize consumption and trading decisions when transactions are costly in proportion to the monetary size of trade.

Portfolio decisions with consumption: two-asset case

In Magill’s and Costantinides’ (1976) model, as well as in many followers, a competitive market in continuous-time is characterized by m risky assets with lognormal price dynamics:

$$\frac{dP_j(t)}{P_j(t)} = \mu_j dt + \sigma_j dZ_j(t) \quad (18.13)$$

and a riskless bank account with constant interest rate r ($m+1$ assets in total):

$$\frac{dP_0(t)}{P_0(t)} = r dt \quad (18.14)$$

Assets are perfectly marketable but transaction costs are incurred in the purchase or sale of each risky security and paid by the bank account.

The latter is increased (decreased) by sales (purchases) and decreased by transaction costs, proportional to the amount exchanged but with possibly different coefficients, k_s , k_b , respectively.

For the decision maker, maximizing his/her lifetime (T-period) expected utility of consumption, the problem is a case of stochastic optimal control and the solution is the optimal trading-consumption policy indicating, at any time t , the buying and selling decisions for each asset and the amount to consume $c(t)$.

A manageable case is obtained assuming just two assets ($m=1$), one risky and one risk-free, and power utility $u(c)=c^{1-\gamma}$, with $CRRA=1-\gamma$.

In contrast to Merton's (1973) result, giving, in absence of transaction costs, a constant optimal asset allocation (myopic trading strategy or Markowitz-Merton portfolio):

$$w^*(t) = w^* = \frac{\mu - r}{(1 - \gamma)\sigma^2} \quad (18.15)$$

and, therefore, a constant risky/riskless asset ratio $w^*/(1-w^*)$, the solution obtained by Magill and Constantinides (1976), amended in Constantinides (1986) and deepened by Davis and Norman (1990) is the identification of a no-trade region (a segment for $m=1$, an area for $m=2$ etc.) between $w_{\min}^*$ and $w_{\max}^*$ so that the optimal trading strategy is:

if $w(t) < w_{\min}^*$ buy the risky asset up to $w_{\min}^*$

if $w(t) > w_{\max}^*$ sell the risky asset down to $w_{\max}^*$

if $w_{\min}^* \leq w(t) \leq w_{\max}^*$ do not trade

This means that the optimal allocation is no longer a point value but an entire region. A graphical representation of the no-trade wedge is in Fig. 18.4.

[Fig. 18.4 about here]

Of course, the critical points $w_{\min}^*$ and $w_{\max}^*$ depend on k_s , k_b and the other parameters of the model.

As reported in Davis and Norman (1990), setting $k_s = k_b = 1.5\%$, $r = 7\%$, $\mu = 12\%$, $\sigma = 40\%$ and $\gamma = -1$ you obtain the Markowitz-Merton allocation $w^* = 15.6\%$ and the no-trade interval ($w_{\min}^* = 9.0\%$, $w_{\max}^* = 19.8\%$).

The optimal trading strategy is to stay inside the interval, preceded by an immediate transaction to the closest point if the initial endowment is outside it (e.g. 100% cash).

As observed by Magill and Constantinides (1976), if there are no transaction costs, the benefit to be gained from improved diversification always induces the investor to continually rebalance the portfolio to the optimal w^* . As soon as the investor is faced with (proportional) transaction costs, this incessant trading would completely dry up the portfolio and he must match the benefits of optimal diversification against the associated transaction costs. The no-trade area identifies the region in which the allocation is not “too distant” from optimal to justify trading or, equivalently, transaction costs would exceed the benefits of improved diversification. The viceversa holds outside this area.

Technically, the optimally controlled portfolio process is a reflecting diffusion inside the wedge and the buying and selling policies, $L(t)$, $U(t)$, represented in monetary terms as cumulative (right-continuous, non decreasing, adapted) processes of all 0 to t purchases and sales (respectively), are the local times of the asset value at the lower (buy) and upper (sell) boundaries of the wedge.

In fact, in absence of transaction costs (the Merton case), the optimization problem can be conditioned to the dynamics of the total wealth process $W(t)$ (budget constraint):

$$\begin{aligned} \frac{dW(t)}{W(t)} &= \left(w(t)\mu + (1 - w(t))r - \frac{c(t)}{W(t)} \right) dt + w(t)\sigma dZ(t) \\ W(t) &\equiv V_1(t) + V_0(t) \equiv P_1(t)Q_1(t) + P_0(t)Q_0(t) \\ w(t) &\equiv \frac{P_1(t)Q_1(t)}{W(t)} \end{aligned} \tag{18.16}$$

$V_j(t)$ being the asset j (monetary) value and $Q_j(t)$ the asset j quantity in t , and the decision variables can be identified in the portfolio allocation $w(t)$ and the consumption rate $c(t)$.

In the case of non-zero transaction costs this representation is no longer possible and we get three decision variables: purchase of the risky asset (= sale of the riskless asset), sale of the risky asset (= purchase of the riskless asset), and consumption.

The representation is now the following:

$$\begin{aligned}dV_0(t) &= (rV_0(t) - c(t))dt - (1 + k_b)dL(t) + (1 - k_s)dU(t) \\dV_1(t) &= \mu V_1(t)dt + \sigma V_1(t)dZ(t) + dL(t) - dU(t)\end{aligned}\tag{18.17}$$

where both consumption and transaction costs deplete the current account, the value of the risky asset is increased by the purchases, $dL(t)$, and decreased by the sales, $dU(t)$ and the triple $(L^*(t), U^*(t), c^*(t))$ (instantaneous control) represents the optimal trading-consumption policy.

Market practice defines $P_1^{\text{ask}} \equiv (1 + k_b)P_1$ as the ask price (buying price) and $P_1^{\text{bid}} \equiv (1 - k_s)P_1$ as the bid price (selling price). Asset quantity at time t is obtained as the initial quantity plus purchases less sales:

$$Q_1(t) = Q_1(0) + \int_0^t \frac{1}{P_1(s)} (dL(s) - dU(s))\tag{18.18}$$

The distinguishing feature of Magill's and Constantinides' (1976) paper is that it reached the substantially correct result without all the required technical devices now available. "The paper was in fact well ahead of its time, in that an essential ingredient of any rigorous formulation, namely the theory of local time and reflecting diffusion, was unavailable to the authors, being at that time (1976) the exclusive property of a small band of pure mathematical votaries. Needless to say, Magill's and Constantinides' paper is far more valuable than many others of unimpeachable mathematical rectitude" (Davies and Norman, 1990).

Clearly, the model is a partial equilibrium. At a more theoretical level, one would wonder if transaction costs admit a general equilibrium in which asset prices are processes of unbounded variation (diffusions or semimartingales) as in the case of no trading costs (for a survey see Amihud, Mendelson and Pedersen, 2005).

With proportional trading costs, even in the simple two-asset case no closed form solutions are available. However, numerical methods, under appropriate conditions on the parameters (e.g.

$0 < r < \mu < r + (1-\gamma)\sigma^2$, can give the unique boundaries $w_{\min}^*$ and $w_{\max}^*$ (see also Dumas and Luciano, 1991).

A graphical representation is provided in Fig. 18.5 as function of the level k ($=k_s=k_b$) of proportional transaction costs.

[Fig. 18.5 about here]

It is interesting to note that as the unit transaction cost k increases, the no-trade area widens but not symmetrically. The upper boundary is less sensitive and the lower boundary decreases more rapidly, so that the optimal no-trade area is shifted toward the riskless asset and the demand for the risky asset is decreasing in the transaction cost rate.

This is an effect of the consumption-investment interaction, given that, by construction, the consumption is withdrawn from the current account. High transaction costs reduce the minimum boundary $w_{\min}^*$ of risky asset given that it may not be worthwhile to buy the asset for investment and later resell it for consumption. Overall, investors accommodate larger transaction costs by reducing the frequency and volume of trade.

In a slightly different model, in which consumption is postponed far in time (e.g. a financial institution), Dumas and Luciano (1991) find both a wider no-trade area and a more symmetric effect of an increasing transaction cost on the two boundaries (see also Taksar, Klass and Assaf, 1988). As they explain, when consumption must be met out of the existing current account, as in Constantinides' model, there is both less room for fluctuations in the amount of cash available and more need to bias the portfolio in favor of cash than when consumption is postponed at a distant future.

An increase of the risk aversion parameter $1-\gamma$ has no significant effect on the width of the no-trade area but it shifts the region toward the riskless asset, given that, as expected, the demand for risky asset is decreasing in risk aversion. The same sensitivities are obtained in case of increasing volatility σ .

Constantinides (1986) also shows that transaction costs have a small decreasing effect on consumption. This is the combination of a positive substitution effect - given that current

consumption is less costly, in term of transaction costs, than future consumption - and a negative income effect - given that higher transaction costs reduce disposable wealth.

Transaction costs have also effect on the equilibrium rate of return of the risky asset. With respect to the case of no cost of trade, a (il)liquidity premium is obtained as compensation for bearing an asset subject to costs of transaction. This illiquidity premium increases with transaction costs and the asset volatility, given that the latter increases the frequency of transactions. According to Constantinides (1986), however, illiquidity premia explained by transaction costs represent only second-order effects on equilibrium asset returns. Therefore, documented, large anomaly like the negative association between average return and market size of stocks (Banz, 1981) cannot be only explained by transaction costs.

This final result has been questioned by Jang, Koo, Liu and Loewenstein (2007) who build a generalized portfolio model with a risky and a risk-free asset but stochastic investment opportunity set.

This means that, at variance with almost all previous literature, the parameters r , μ , σ are stochastically changing and optimal trading policies significantly affected.

In particular, they assume two regimes, “Bull”, B , and “bear”, b , so that all the fundamental parameters are regime-dependent and the financial market switches from one regime into the other at the first jump of a Poisson process with state dependent switching probability. Given that $\sigma_B < \sigma_b$ an investor can infer the state of the current regime from observing stock return volatility.

One relevant result is that not only the optimal trading strategy with proportional transaction costs is no longer myopic, with two – possibly overlapping – no-trade regions, one for Bull and one for bear regime, but also that the trading strategy in one regime is affected by the investment opportunity set in the other regime, so that the investor smooths not only consumption but also trading strategies across regimes. For example, using typical parameter values (e.g. $\sigma_B = 13\%$, $\sigma_b = 26\%$, Bull average duration 4.25 years, bear average duration 0.58 years) the investor tends to hold more risky asset in the bear regime (with respect to the no transaction cost case) to reduce transaction costs upon regime switching, given the relative shorter duration of bear with respect to Bull phase. The effect is increased if Bull proportional costs are higher than bear costs: it is

cheaper, in this case, to buy the asset in the bear regime – which is expected to last a short time – and enjoy the benefits in the Bull regime.

Portfolio decisions with consumption: many-asset case

The generalization of the allocation model with transaction costs to $m > 1$ risky assets presents, in general, exponentially increasing complexities. In fact, j (between 0 and m) risky securities can be

bought or sold in 2^j ways and j securities can be chosen from m in $\binom{m}{j} \equiv \frac{m!}{j!(m-j)!}$ ways, so that

there are $\binom{m}{j} 2^j$ different regions in which only j securities are transacted in all possible ways.

Given that j goes from 0 to m the total number of possible regions is $\sum_{j=0}^m \binom{m}{j} 2^j = \sum_{j=0}^m \binom{m}{j} 2^j 1^{m-j} = 3^m$, i.e 3 for $m=1$ (Buy, No-trade, Sell), 9 for $m=2$ (BB,BN,BS,NB,NN,NS,SB,SN,SS),, 59049 for $m=10$.

Notwithstanding these difficulties, Akian, Menaldi and Sulem (1996) provide a multi-dimensional version of Davis and Norman (1990) assuming uncorrelated assets and Liu (2004) include both fixed and proportional transaction costs.

The presence of fixed costs changes the optimal strategy by including discrete, lump-sum trades (impulse control).

Assuming homogeneous costs across assets, the current account dynamics are now:

$$dV_0(t) = (rV_0(t) - c(t))dt - \sum_{j=1}^m \left[(1 + k_b) dL_j(t) + (1 - k_s) dU_j(t) + K_b 1_{dL_j > 0} + K_s 1_{dU_j > 0} \right]$$

where $1_{A>0}$ is the indicator function of the event $\{A>0\}$ and K_b (K_s) is the fixed cost per buy (sale) trade.

The optimal trading strategy involves, for each asset j , four critical points: $w_{j,min}^* < w_{j,min}^{**} < w_{j,max}^{**} < w_{j,max}^*$, and can be described as follows:

if $w_j(t) \leq w_{j,min}^*$ then buy asset j to jump to $w_{j,min}^{**}$

if $w_j(t) \geq w_{j,max}^*$ then sell asset j to drop to $w_{j,max}^{**}$

if $w_{j,\min}^* < w_j(t) < w_{j,\max}^*$ then do not trade

The result is depicted in Fig. 18.6. As long as the portfolio allocation lies within the outer, signaling boundaries, no trade is made. When it hit one of the boundaries a lump sum trade is made toward the nearest inner boundary

[Fig. 18.6 about here]

As the transaction costs increase, the no-trade area widen and the optimal transaction frequency decreases, in particular for buy decisions. Using reasonable estimates (5\$ for fixed and 1% for proportional transaction costs), Liu (2004) finds, on average, one optimal sale per year and one optimal purchase in 2.5 years, so that monthly or daily trade frequency would induce a substantial utility loss. The potential net gains from stock picking, market timing and return predictability (e.g. Black and Litterman, 1992, Kandel and Stambaugh, 1996, Barberis, 2000) should accordingly be revised.

Note, however, that the level of transaction costs is asset specific, given that different assets have different levels of illiquidity. In general, whenever futures markets are available for certain securities or asset classes, the level of transaction costs could be significantly reduced (Clarke, 1992).

Note also that even if the proportional cost k is 0, even a small fixed transaction cost K generates a no-trade area which makes the Merton allocation w^* no longer optimal to keep continuously. In this case, Merton allocation w^* is essentially the target to jump to whenever the current portfolio pierces the boundaries. Note also that as the proportional cost k increases, both the two upper and the two lower boundaries converge, meaning that the size of a sale (upper case) or purchase (lower case) decreases.

One interesting result of Liu (2004) model is that, notwithstanding the multiplicity of assets, the investor has to trade, with probability 1, at most in one asset at a time. This is not the effect of the assumption of uncorrelated assets but a more general effect of the not perfect substitutability of one asset for any other. Increasing the degree of substitutability (i.e. correlation) among assets,

there is a reduction of investment in each asset because of a smaller diversification benefit. Moreover, in contrast with Constantinides' results, the average, long-run allocation in risky assets is increased by non zero transaction costs and a greater asset volatility σ_j lowers the boundaries and narrows the no-trade area. This is because the greater risk reduces the asset optimal holdings and increases its monitoring and rebalancing frequency. The opposite holds in the case of a greater expected return μ_j , which increases the boundaries and widens the no-transaction region.

Portfolio decisions with trading costs and absolute targets

It is straightforward to generalize the Markowitz (1952, 1959) –Tobin (1958) mean-variance one-period portfolio model to the case of transaction costs.

The maximizing target is total portfolio return ($m+1$ assets) adjusted for risk (the covariance matrix Σ) and for transaction costs, measured as squared function of the distance between the optimal risky portfolio w and the inherited portfolio w_0 :

$$\max_w \left\{ w' \mu + (1 - w' \mathbf{1})r - \frac{\vartheta}{2} w' \Sigma w - \frac{1}{2} (w - w_0)' \Lambda (w - w_0) \right\} \quad (18.19)$$

The parameter ϑ represents the investor's aversion to risk and the vector $\Lambda/2(w-w_0)$ (with Λ possibly diagonal and related to Σ) represents the transaction costs of exchanging w_0 for w (see Gârleanu and Pedersen, 2011 for a micro-foundation).

The first order conditions give the optimal portfolio:

$$w^* = (\vartheta \Sigma + \Lambda)^{-1} (\mu - r \mathbf{1} + \Lambda w_0) \quad (18.20)$$

showing that transaction costs have the joint effect of an increase of variance for all risky assets and an increase of expected return for all inherited positions. Markowitz portfolio is recovered by putting $\Lambda=0$.

Assuming $\Lambda=\lambda\Sigma$, so that transaction costs (bid-ask spread) increase with market volatility (cf. Roll, 1984) you obtain the simple result that w^* is a weighted average of the classical Markowitz portfolio (with no transaction costs) and the old portfolio:

$$w^* = \frac{\lambda}{\vartheta + \lambda} w_0 + \frac{\vartheta}{\vartheta + \lambda} (\vartheta \Sigma)^{-1} (\mu - r \mathbf{1}) = w_0 + \frac{\vartheta}{\vartheta + \lambda} (w_{\text{Mark}} - w_0) \quad (18.21)$$

The portfolio rebalancing is shown in Fig. 18.7, indicating that the optimal portfolio is toward the Markowitz portfolio in proportion of risk aversion ϑ and it is toward the old position in proportion to transaction costs λ .

[Fig. 18.7 about here]

An intertemporal discrete-time extension can be obtained, following Gârleanu and Pedersen (2011), assuming that the investor is willing to choose the dynamic trading strategy $(w_0, w_1, \dots, w_t, \dots)$ to maximize the expected value of all discounted mean-variance utilities:

$$\max_{w_0, w_1, \dots} E_0 \left[\sum_{t=0}^{\infty} (1-\rho)^t \left(r + w_t' \eta_t - \frac{\vartheta}{2} w_t' \Sigma w_t - \frac{1}{2} (w_t - w_{t-1})' \Lambda (w_t - w_{t-1}) \right) \right] \quad (18.22)$$

Note that ρ is the time-preference discount rate, Σ and Λ are constant symmetric matrices of covariances and transaction costs respectively, and the excess return vector $\eta_t \equiv \mu_t - r1$ is assumed a stationary mean reverting process according to the dynamics:

$$\eta_{t+1} - \eta_t = \Phi(\eta - \eta_t) + \varepsilon_{t+1} \quad (18.23)$$

The intertemporal optimization gives, as before, an intuitive results: the optimal portfolio is a “matrix-weighted average” of previous portfolio w_{t-1} and a dynamic-target portfolio z_t :

$$w_t = (I - A)w_{t-1} + Az_t \quad (18.24)$$

so that the optimal trade is $w_t - w_{t-1} = A(z_t - w_{t-1})$ for an optimal matrix A , nonlinear function of ρ , ϑ , Σ , and Λ .

Equivalently, w_t is a weighted average of previous portfolio, Markowitz portfolio and the expected value of next target:

$$w_t = Bw_{t-1} + C(\vartheta\Sigma)^{-1}\eta_t + (I - B - C)E_t(z_{t+1}) \quad (18.25)$$

If $\Lambda = \lambda\Sigma$ the solution becomes simpler:

$$\begin{aligned}
w_t &= \left(1 - \frac{a}{\lambda}\right) w_{t-1} + \frac{a}{\lambda} z_t \\
&= \frac{\lambda}{\lambda + \vartheta + (1-\rho)a} w_{t-1} + \frac{\vartheta}{\lambda + \vartheta + (1-\rho)a} (\vartheta \Sigma)^{-1} \eta_t + \frac{(1-\rho)a}{\lambda + \vartheta + (1-\rho)a} E_t(z_{t+1}) \\
z_t &= (\vartheta \Sigma)^{-1} \left(I + \frac{a(1-\rho)}{\vartheta} \Phi \right)^{-1} \eta_t \\
a &= \frac{-(\vartheta + \lambda\rho) + \sqrt{(\vartheta + \lambda\rho)^2 + 4\vartheta\lambda(1-\rho)}}{2(1-\rho)}
\end{aligned} \tag{18.26}$$

The first line states that the optimal portfolio is a balance between the previous portfolio w_{t-1} and an optimal target z_t , with a weight, a/λ , on the latter which decreases in trading costs λ , because higher trading costs imply a penalty to leaving the inherited portfolio, and increases in risk aversion ϑ , because higher risk aversion means a penalty to straying too far from the optimal target z_t . Note also that as ρ increases to 1 the investor is increasingly impatient, discounting heavily the future up to the limit case (a goes to $\vartheta\lambda/(\vartheta+\lambda)$) in which the optimal portfolio is the (myopic) static portfolio w^* .

The second line provides an equivalent decomposition involving not only the previous portfolio w_{t-1} but also the Markowitz portfolio, optimal under zero trading costs, and the expected value of the future target portfolio z_{t+1} . The optimal rebalancing is depicted in Fig. 18.8.

[Fig. 18.8 about here]

One interesting version of Gârleanu and Pedersen (2011) model is obtained in their main case of predictable returns. A vector of predictors, f_t , stationary and zero-mean-reverting, representing the private information of the investor (or asset manager), helps forecast the asset returns so that:

$$\begin{aligned}
\mu_t &= \mu + Bf_t \\
f_{t+1} - f_t &= -\Phi f_t + \varepsilon_{t+1}
\end{aligned} \tag{18.27}$$

The matrix B includes the predictor loadings and the vector $\alpha_t \equiv Bf_t$ is the vector of active “alphas” of the asset manager, whose objective is to maximize the expected value of all discounted portfolio’s alphas $w_{t+h}'\alpha_{t+h}$ adjusted for portfolio risk and transaction costs. Given that the alphas are expected to go to zero in the long run, the optimal Markowitz portfolio is expected to converge to the steady-state vector $(\vartheta\Sigma)^{-1}\mu$ and actual portfolios are optimally driven to exploit the available alphas, taking into account, as seen before, portfolio risks and transaction costs. This expected evolution is depicted in Fig. 18.9. See Gârlenau and Pedersen (2011) for a continuous-time version of their model.

[Fig.18.9 about here]

Portfolio decisions with trading costs and benchmark

From the point of view of asset managers, portfolio decisions could be conveniently set out assuming an exogenous, fixed target portfolio allocation w^* (CM strategic portfolio), which represents the benchmark reference in the mandate of the investor (principal) to the fund manager (agent).

As asset values move randomly, actual portfolio $w(t)$ diverges from the target w^* and the asset manager has to optimally drive the portfolio composition taking into account both tracking error costs, depending on $w(t)-w^*$, and trading costs, depending on $|w(t+1)-w(t)|$. The tracking error is usually measured by the variance (or the standard deviation) of the return difference between current and optimal portfolio. Following Leland (1999), both tracking error and transaction costs create utility loss and the optimal investment policy will minimize the present value of tracking error costs plus transaction costs over the mandate maturity T :

$$\min_w E_t \left\{ \int_t^T e^{-r(s-t)} \vartheta (w(s) - w^*)' \Sigma (w(s) - w^*) ds + \int_t^T e^{-r(s-t)} k' |dw(s)| \right\} \quad (18.28)$$

where ϑ is the parameter representing the manager’s “tracking error aversion” and k' the vector of assets trading costs. Note that all costs are measured per unit of wealth and the objective function

is consistent with expected utility maximization for the instantaneous net utility $U(w^*) - \vartheta(w(s) - w^*)' \Sigma(w(s) - w^*) - k' |dw(s)|$.

Alternatively, as in Sun, Fan, Chen, Schouwenaars and Albota (2006) and Kritzman, Myrgren and Page (2009), the tracking error costs for not being at the optimal allocation w^* could be measured as certainty-equivalent loss (in dollar terms) between $w(t)$ and w^* .

It should not be surprising that Leland (1999) (approximated) results, obtained in the infinite horizon case ($T=\infty$), are qualitatively similar to Magill and Constantinides (1976). The optimal policy is represented by a connected, compact no-trade region expressed, in the riskless-risky two asset case by the segment $[w_{\min}, w_{\max}]$ where $w_{\min}$ and $w_{\max}$ are obtained as solution of the (ordinary) differential problem:

$$\begin{aligned} \frac{1}{2} w^2 (1 - w^*)^2 \sigma^2 J_{ww} + w(1 - w^*)(\mu - r - \sigma^2 w^*) J_w + \vartheta \sigma^2 (w - w^*)^2 - rJ &= 0 \\ J_w|_{w=w_{\min}} = -J_w|_{w=w_{\max}} &= -k \quad (\text{value matching conditions}) \\ J_{ww}|_{w=w_{\min}} = -J_{ww}|_{w=w_{\max}} &= 0 \quad (\text{super contact conditions}) \end{aligned} \quad (18.29)$$

where $J(w(t))$ is Bellman's value function.

A numerical result with $m=2$ risky assets is given in Fig. 18.10 for $w^*=(40\%,40\%)$.

[Fig. 18.10 about here]

The upper panel compares the no-trade regions when the two assets have low (20%) or high correlation (70%). Increasing the correlation increases the skewness of the region given that the two assets are closer substitute in the latter case. In the lower panel, the reduction in trading costs for asset 2 (from 1% to 0.1%) shrinks the no-trade region in that dimension but also, due to the correlation effect, moves the optimal boundaries of the more expensive asset 1: more (less) of asset 2 implies less (more) of asset 1.

Another interesting result of Leland's (1999) model is that it provides an estimate of the effects of a "Tobin tax": a 2% tax on portfolio trades in addition to 0.5% of usual transaction costs would reduce trading volume by about 42%. Moreover, using optimal rebalancing would reduce

turnover by about 50% with respect to sub-optimal rules like periodic rebalancing with the same tracking error volatility: a significant reduction in costs without any loss in controlling the risks.

Simple rebalancing rules: a comparison

A quick review of the literature (Arnott and Lovell, 1993, Buetow, Seller, Trotter, Hunt and Whipple, 2002, Masters, 2003, O'Brien, 2006) shows that a number of simple rules of portfolio rebalancing have been proposed and used in actual practice of portfolio managers.

As documented by Cao (2009) and Hong (2009), individual long-term investors are often advised to follow the rule of thumb of checking at birthday (i.e. once a year) and rebalancing if the portfolio deviates from target by more than 5%, given that “letting a portfolio untended is like leaving a toddler alone in a room with a hot stove” (Citigroup, 2005).

In addition to the risk reduction motivation, rebalancing is often suggested as a tool for yield enhancement given that, in its usual meaning, it implies the trading rule “buy low - sell high” (contrarian strategy) which, under price mean-reversion, will improve portfolio performance (Perold and Sharpe, 1988, Cesari and Cremonini, 2003). This is sometimes referred to as the “rebalancing bonus”.

The need for rebalancing depends on the asset characteristics and in particular (Pliska and Suzuki, 2004):

- directly on volatility, expected return and time because they imply high probability of large deviations of one asset against the others;
- inversely on correlations because high correlations indicate high asset co-movements and less need for rebalancing. For example, more specialized portfolios (e.g. 90/10) should require less rebalancing than mixed ones (e.g. 50/50).

Given a benchmark allocation, w^* , and a pre-specified periodic monitoring, some popular rules are the following:

- time rebalancing: rebalance the portfolio to w^* every given time period (day or week, month, quarter, year);
- fixed rebalancing: rebalance when $|w_j(t) - w_j^*| > a\%$

- proportional rebalancing: rebalance when $|w_j(t) - w_j^*| / w_j^* > b\%$
- probability rebalancing: rebalance when $|R_j(t_r, t)| > c\% \sigma_j(t)$, where $R_j(t_r, t)$ is the annualized rate of return of asset j between the last rebalancing date t_r and current date t and $\sigma_j(t)$ is the estimated annualized volatility up to time t . In this case the trigger is an “excess return” (in either direction) for asset j , independently from its current or optimal weight.

Note that time rebalancing (e.g. monthly) is equivalent to periodic (monthly) monitoring and fixed rebalancing with $a\% = 0\%$. BH, instead, is fixed rebalancing (or other rule) with $a\% = 100\%$ so that rebalancing will never occur.

Note also that different rules are obtained by combining different monitoring periods with different trigger parameters $a\%$, $b\%$, $c\%$. Moreover, instead of full rebalancing to w_j^* , a rebalance to $w_j^* \pm a\%$ (rebalance boundary) or to $w_j^* \pm q\% a\%$ (tolerance boundary) could be implemented whenever $|w_j(t) - w_j^*| > a\%$ (idem for the other rules).

Notwithstanding the available, heterogeneous evidence, some general results can be grasped from the quoted empirical analyses:

- historical simulation shows that, on average, rebalancing activity provides extra returns (net of transaction costs) with respect to BH, with important exception in periods of sufficiently strong up- or down- market trends;
- when market regime cannot be recognized, higher monitoring frequency is required;
- with high frequency monitoring (daily, weekly) the best results are obtained with large boundaries (e.g. $a\% = 15\%$) but with a low frequency monitoring (yearly) narrow boundaries (e.g. 1%) are to be preferred;
- on average, fixed rebalancing and proportional rebalancing provide superior returns (also adjusted for risk) with respect to BH as well as the popular time rebalancing. Their resemblance to the optimal theoretical rebalancing policy could explain this result.

Portfolio decisions with monitoring costs

Monitoring activity is usually considered costless. In practice, however, it involves a number of operations and actions (info collecting, price monitoring, market analysis, back-office reporting) that

are time and resource consuming and therefore costly. If these activities can be switched on/off it may be optimal to discretize monitoring and endogenize its time frequency.

In a recent paper, Bukhvalova (2007) analyses, in absence of transaction costs, portfolio decisions with monitoring costs, defined as fixed proportion v of beginning of period wealth,. Her results show that over the given horizon T , there is an optimal number N^* of periods (depending on v , r and T) of equal length in which the investor has to monitor the portfolio. The N^* is increasing with T and r and decreasing in the unit cost v . For example, with a monitoring cost of 4 bps, $r=2\%$ and an horizon of 20 years you obtain 5 rebalancing periods. Portfolio allocation is affected as well and the optimal weight $w^*(t)$ is time dependent and below the zero-cost case in previous periods, above it in final periods.

TARGET REBALANCING FOR PORTFOLIO PROTECTION

We have defined target rebalancing as the dynamic adjustment of strategic portfolio weights $w^*(t)$.

Two kinds of target rebalancing are interesting to be mentioned because they are both aimed at protecting the future value of the portfolio. One is portfolio insurance (Black and Jones, 1987, Black and Perold, 1992) and the other one is life-cycle investment (Campbell and Viceira, 2002, ch.7).

Portfolio insurance

Portfolio insurance is an automatic rebalancing strategy trying to “guarentee” (in a probabilistic, non-contractual sense) to reach at least a final level $F(T)$ (floor) for the portfolio value at time T .

In simple terms, the portfolio is composed by a riskless (current account) and a risky asset and the portfolio composition must be periodically adjusted in order to have a reasonable certainty to reach the minimum target $F(T)$.

Let $V(0)$ be the initial wealth and $F(0)=F(T) e^{-rT}$ be the initial value of the floor. The simple rule suggested by Black and Jones (1987) is to rebalance the risky position, in monetary terms, to:

$$E(t) = \begin{cases} m(V(t) - F(t)) & \text{if } V(t) > F(t) \\ 0 & \text{if } V(t) \leq F(t) \end{cases} \quad (18.30)$$

The parameter $m > 1$ is called the multiplier and $C(t) \equiv V(t) - F(t)$ is called the cushion so that the rule (constant proportion portfolio insurance, CPPI) could be expressed *à la Einstein* as $E = mC$. It means that the value of the risky asset must be in proportion (at level m) with the distance between the portfolio value $V(t)$ and the floor value $F(t)$. The greater (shorter) the distance, the greater (lower) the risky exposure.

If $P(t)$ is the unit price of the risky asset and $P_0(t) = e^{rt}$ is the unit value of the current account, the optimal quantity are, therefore:

$$Q^*(t) = \frac{m(V(t) - F(t))}{P(t)} \quad Q_0^*(t) = \frac{V(t) - m(V(t) - F(t))}{e^{rt}} \quad (18.31)$$

and the budget constraint is:

$$V(t) = Q(t)P(t) + Q_0(t)e^{rt} = Q^*(t)P(t) + Q_0^*(t)e^{rt} \quad (18.32)$$

so that, being the portfolio self-financing:

$$dV(t) = Q^*(t)dP(t) + Q_0^*(t)dP_0(t) = \frac{m(V(t) - F(t))}{P(t)}dP(t) + \frac{V(t) - m(V(t) - F(t))}{e^{rt}}e^{rt}rdt \quad (18.33)$$

The CPPI rule (18.30) implies, therefore, a general solution for $V(t)$ of the form:

$$V(t) = F(t) + hP^m(t)e^{(1-m)rt} \quad (18.34)$$

showing that the NAV of the portfolio (portfolio payoff) is a convex function of the price $P(t)$ of the risky asset. Note that for $0 < m < 1$ and $F(0) = 0$ the NAV is concave and we have the CM strategy.

This convexity, obtained with $m > 1$, is a typical effect of the insurance rebalancing behavior. It can be found, under apparently different forms, in many situation in which the investor or the portfolio is intentionally hedging against adverse movement of the market. Two well known examples are the market timing activity of the asset manager (Treynor and Mazuy, 1966 and Henriksson and Merton, 1981) and the protective put option, both characterized by typical convex payoff. In fact, similar results are obtained when the protective rebalancing is defined by the dynamic replication of a put option with strike $F(T)$ (option based portfolio insurance, see Cesari and Cremonini, 2003).

A second feature of protective rebalancing (and insurance policies in general) is its “momentum” character. In fact putting (18.34) into (18.31) we obtain:

$$\frac{\partial Q^*(t)}{\partial P(t)} = (m-1) \frac{Q^*(t)}{P(t)} > 0 \quad \text{for } m > 1$$

so that the rule is buy (sell) if the risky price goes up (down). This means that with portfolio insurance you go with the market, “buy high - sell low”, following the market momentum. In fact, a generalized adoption of portfolio insurance rules by market participants could substantially increase market volatility and the probability of booms and crashes, as documented for the “Black Monday”, in October, 19th, 1987 (Rubinstein, 1988).

In actual practice (Etzioni, 1986), discrete portfolio rebalancing cannot guarantee the final floor $F(T)$ and the risk of default for the strategy is a direct function of the multiplier m , the price volatility σ and the time distance from subsequent rebalancing.

Life-cycle investments

A second example of target rebalancing for portfolio protection is given by life-cycle investments.

Life-cycle investment is, in fact, a popular expression used to indicate a dynamic asset allocation for long term investors in which, as long as a given final date is approaching (typically the retirement date), the investor is advised/committed to progressively reduce his or her exposure to risky assets (stocks, in particular) in favor of safer, short term bonds, until the end date is reached with a low-risk portfolio. The stated goal for such an age-based asset allocation is twofold: to take profit from the long-term growth of stocks but also to preserve, over the lifetime, the investor's accumulated wealth, avoiding, in particular, the unpleasant, last moment surprises from adverse market movements.

In practice, the optimal portfolio weights $w^*(t)$ for risky assets are gradually decreased with the investor approaching the retirement date T . A colorful rule of thumb says $w^*(t)=100-\text{age}(t)$, suggesting a risky exposure of about 80% for the young and 20% for the elder. This could be done with periodic switches from higher-risk into lower-risk mutual funds or investing in the same target-date fund with time-decreasing risk.

In general, we have life-cycle investments whenever there is an “horizon effect” on the optimal portfolio, i.e. $w^*(t)$ is a direct function of $T-t$.

The rationale supporting the strategy is in a few “postulates” which fed a long and lively debate: the risk premium of stocks, their reduced volatility in the long-run because of mean-reversion (so called “time-diversification” effect), the human capital as personal wealth (see Viceira, 2007 and Cesari, 2011).

The actual ability of life-cycle investing to provide excess returns and risk reduction in the long run, against for example, the more traditional CM is still under active scrutiny (e.g. Soto, Triest, Golub-Sass, Golub-Sass, 2008).

SUMMARY AND CONCLUSIONS

Both active and passive asset management imply adequate trading strategies, suitable defined in terms of goals and constraints. The intrinsic randomness of financial markets suggests periodic monitoring and rebalancing of the portfolio in order to be fully compliant with internal and external rules and in line with expectations and targets.

Even in the case of classical, static portfolio results (Markowitz, 1952, 1959), valuable - under special assumptions - also in a dynamic context (Merton, 1969, 1971, 1973), there is scope to rebalancing activity due to the random movements of relative asset prices.

In general, in addition to passive rebalancing, it is possible to define a target rebalancing and a portfolio rebalancing toward target. In fact, intertemporal models show that the target itself could be time-dependent, enlarging the scope for a fine-tuning of portfolio weights. Useless to say, it is the presence of transaction costs making optimal rebalancing a major issue.

The path-breaking work of Magill and Constantinides (1976) and Constantinides (1986) has shown the upsetting effects of transaction costs on rebalancing activity and portfolio-consumption decisions. In their model, the optimal portfolio is replaced by optimal boundaries in the space of asset weights, defining a connected no-trade region: if current portfolio pierces the boundaries, an instantaneous trade is made to rebalance the portfolio at (case of proportional costs) or inside (case of fixed costs) the boundaries. In the case of many risky assets, the rebalancing problem is extremely challenging and only numerical solutions are available (Liu, 2004).

From a management point of view, an interesting set up for rebalancing decisions is obtained in the case of a given target w^* , from which actual portfolio strays away by random market movements (Leland, 1999). The asset manager has to trade-off the costs of erroneously tracking the target and the costs of trading it against current positions. Even in this case a no-trade region defines the optimal policy.

In actual practice, financial advisors and fund managers suggest a number a simple, rebalancing rules with the aim of both controlling the risk and enhancing the return of the portfolio. Periodic rebalancing or fixed-proportional rebalancing are the most popular ones, implying a “buy low – sell high” trading rule (contrarian strategy). It can be shown, however, both theoretically and empirically, that they cannot guarantee under all market regimes (no free lunch in the market) positive excess returns with respect to the basic “buy and hold” rule.

Similar results hold for the opposite strategies (momentum strategies) often adopted or suggested by fund managers. Both portfolio insurance and life-cycle investments try to protect the portfolio using a “buy high – sell low” rule, inducing convexity in the allocation to hedge the accumulated wealth from downside risks. The actual implementation of these strategies jeopardizes their theoretical optimality so that their risk/return efficiency with respect to more traditional BH or CM strategies is controversial.

APPENDIX: THE ALGEBRA OF PORTFOLIO DYNAMICS

By definition, the (net asset) value of a portfolio (NAV) at time t is the sum of quantities times prices of all assets (and liabilities with negative sign, if this is the case):

$$V(t) \equiv \sum_{j=1}^N Q_j(t) P_j(t) \quad (A.1)$$

The change in value is given by:

$$\begin{aligned} V(t+1) - V(t) &= \sum_{j=1}^N Q_j(t+1) P_j(t+1) - \sum_{j=1}^N Q_j(t) P_j(t) \\ &= \sum_{j=1}^N [Q_j(t+1) - Q_j(t)] P_j(t+1) + \sum_{j=1}^N Q_j(t) [P_j(t+1) - P_j(t)] \end{aligned} \quad (A.2)$$

where the first addendum in the last equality in (A.2) is the (new) investment and the second one is the capital gain/loss from the (old) time- t portfolio. Note that investment (disinvestment if negative) is given by the change in the quantities of assets in the portfolio.

By the price taking assumption, in fact, investors can control the quantities, not the prices, adding up to the portfolio value.

A trading strategy is a rule governing the dynamics of the asset quantities.

Moreover, investment must satisfy the balance constraint, stating that uses and sources of value must be equal:

$$\sum_{j=1}^N Q_j(t+1) P_j(t+1) + \text{outflow} = \sum_{j=1}^N Q_j(t) P_j(t+1) + \sum_{j=1}^N Q_j(t) D_j(t+1) + \text{inflow} \quad (A.3)$$

where $D_j(t+1)$ is the dividend (or coupon) income per share obtained from asset j between t and $t+1$.

Equivalently, investment must be equal to saving:

$$\sum_{j=1}^N [Q_j(t+1) - Q_j(t)] P_j(t+1) = \sum_{j=1}^N Q_j(t) D_j(t+1) + \text{net inflow} \quad (A.4)$$

Substituting in (A.2) we have:

$$\begin{aligned}
V(t+1) - V(t) &= \sum_{j=1}^N Q_j(t) [P_j(t+1) - P_j(t)] + \sum_{j=1}^N Q_j(t) D_j(t+1) + \text{net inflow} \\
&= \sum_{j=1}^N Q_j(t) P_j(t) \left[\frac{P_j(t+1) - P_j(t)}{P_j(t)} + \frac{D_j(t+1)}{P_j(t)} \right] + \text{net inflow} \quad (\text{A.5})
\end{aligned}$$

The amount in square brackets is the **one-period total rate of return** of asset j , $R_j(t+1)$, sum of the capital gain/loss $R_j^c(t+1)$ and the dividend/coupon yield $R_j^d(t+1)$ of the asset.

If net inflow is zero the portfolio is **self-financing** and dividends are reinvested in the portfolio.

$$V(t+1) - V(t) = \sum_{j=1}^N Q_j(t) [P_j(t+1) - P_j(t) + D_j(t+1)] \quad (\text{A.6})$$

Comparing this equation with (A.2) you see that in a self-financing portfolio, once prices have been adjusted for dividends, the change in quantities can be set to zero.

Assuming, in the following, that this is the case we obtain:

$$R(t+1) \equiv \frac{V(t+1) - V(t)}{V(t)} = \sum_{j=1}^N \frac{Q_j(t) P(t)}{V(t)} R_j(t+1) \equiv \sum_{j=1}^N w_{j,t,t} R_j(t+1) \quad (\text{A.7})$$

where $w_{j,t,t}$ is the percentage value-weight of asset j (first suffix) in the portfolio using quantities (second suffix) and prices (third suffix) at time t .

The T -period wealth is:

$$\begin{aligned}
V(T) &= V(0) + \sum_{t=0}^{T-1} (V(t+1) - V(t)) = V(0) + \sum_{t=0}^{T-1} V(t) R(t+1) \\
&= V(0) \prod_{t=0}^{T-1} \frac{V(t+1)}{V(t)} = V(0) \prod_{t=0}^{T-1} (1 + R(t+1)) \quad (\text{A.8})
\end{aligned}$$

where the first line provides an additive composition and the second line the (more useful) factorial composition. Using (A.7) we have:

$$V(T) = V(0) \prod_{t=0}^{T-1} \sum_{j=1}^N w_{j,t,t} (1 + R_j(t+1)) \quad (\text{A.9})$$

The T -period total return, between 0 and T is:

$$\begin{aligned}
1 + R(T) &\equiv \frac{V(T)}{V(0)} = \frac{\sum_{j=1}^N \frac{Q_j(T)P_j(T)}{\sum_{j=1}^N Q_j(0)P_j(0)}}{\sum_{j=1}^N \frac{Q_j(0)P_j(0)}{\sum_{j=1}^N Q_j(0)P_j(0)}} = \sum_{j=1}^N \frac{Q_j(0)P_j(0)}{\sum_{j=1}^N Q_j(0)P_j(0)} \frac{Q_j(T)P_j(T)}{Q_j(0)P_j(0)} \\
&= \sum_{j=1}^N w_{j,0,0} \prod_{t=0}^{T-1} (1 + q_j(t+1))(1 + R_j^c(t+1)) \\
&= \prod_{t=0}^{T-1} \sum_{j=1}^N w_{j,t,t} (1 + R_j(t+1))
\end{aligned} \tag{A.10}$$

where the last equality is from (A.9).

Note that $q_j(t+1)$ is the rate of growth of the quantity of asset j between t and $t+1$, financed by dividends and sales of other assets.

One possibility is the purchase of shares of asset j using its dividends. This implies:

$$Q_j(t+1) = Q_j(t) + \frac{Q_j(t)D_j(t+1)}{P_j(t+1)}$$

or

$$1 + R_j(t+1) \equiv 1 + R_j^c(t+1) + R_j^d(t+1) = (1 + q_j(t+1))(1 + R_j^c(t+1)) \tag{A.11}$$

An alternative, equivalent representation is the following:

$$\begin{aligned}
1 + R(T) &\equiv \frac{V(T)}{V(0)} = \prod_{t=0}^{T-1} \frac{V(t+1)}{V(t)} = \prod_{t=0}^{T-1} \sum_{j=1}^N \frac{Q_j(t+1)P_j(t+1)}{\sum_{j=1}^N Q_j(t)P_j(t)} \\
&= \prod_{t=0}^{T-1} \sum_{j=1}^N w_{j,t,t} (1 + q_j(t+1))(1 + R_j^c(t+1))
\end{aligned} \tag{A.12}$$

Clearly, the trading strategy, i.e. the rule governing $q_j(t+1)$, will affect the T -period return, its level and volatility.

Buy and hold trading strategy

A buy-and-hold strategy (BH) in the no-dividend case is simply given by $Q_j(T)=Q_j(0)$ so that:

$$\begin{aligned}
1 + R_{BH}(T) &= \frac{\sum_{j=1}^N \frac{Q_j(0)P_j(T)}{\sum_{j=1}^N Q_j(0)P_j(0)}}{\sum_{j=1}^N \frac{Q_j(0)P_j(0)}{\sum_{j=1}^N Q_j(0)P_j(0)}} = \sum_{j=1}^N \frac{Q_j(0)P_j(0)}{\sum_{j=1}^N Q_j(0)P_j(0)} \frac{P_j(T)}{P_j(0)} = \sum_{j=1}^N w_{j,0,0} \frac{P_j(T)}{P_j(0)} \\
&= \sum_{j=1}^N w_{j,0,0} \prod_{t=0}^{T-1} (1 + R_j^c(t+1))
\end{aligned} \tag{A.13}$$

Alternatively, from (A.12):

$$\begin{aligned}
1 + R_{BH}(T) &\equiv \frac{V(T)}{V(0)} = \prod_{t=0}^{T-1} \frac{V(t+1)}{V(t)} = \prod_{t=0}^{T-1} \sum_{j=1}^N \frac{Q_j(0)P_j(t+1)}{\sum_{j=1}^N Q_j(0)P_j(t)} \\
&= \prod_{t=0}^{T-1} \sum_{j=1}^N w_{j,0,t} (1 + R_j^c(t+1))
\end{aligned} \tag{A.14}$$

Assuming that dividends are non zero and are invested in the same portfolio, to buy shares of the same asset we have the following adjustment, where dividends, $Q_j(t)D_j(t+1)$, are used to buy $Q_j(t)D_j(t+1)/P_j(t+1)$ new shares of asset j :

$$1 + R_{BH}(T) = \sum_{j=1}^N \frac{\left[Q_j(0) + \sum_{t=0}^{T-1} Q_j(t) \frac{D_j(t+1)}{P_j(t+1)} \right] P_j(T)}{\sum_{j=1}^N Q_j(0)P_j(0)} \tag{A.15}$$

Note that a number of well known benchmark indices (provided by Standard & Poor, Dow Jones, Morgan Stanley and many others) are – a part from entries, exits and other basket adjustments - a sort of BH value-weighted portfolios. Note that portfolio rebalancing must not be confused with so called index rebalancing. According to Siegel and Schwartz (2006), the latter typically reduces index performance due to positive momentum of exclusions and poor long-term returns of inclusions.

Constant mix trading strategy

A constant-mix strategy (CM) is defined as a relative TS whose goal is to keep fix the portfolio proportions.

This means:

$$w_{j,t+1,t+1} \equiv \frac{Q_j(t+1)P_j(t+1)}{\sum_{h=1}^N Q_h(t+1)P_h(t+1)} = \frac{Q_j(t)P_j(t)}{\sum_{h=1}^N Q_h(t)P_h(t)} \equiv w_{j,t,t} = w_{j,0,0} \quad \text{for all } t \tag{A.16}$$

so that rebalancing is obtained according to the CM rule:

$$Q_j(t+1) = \frac{Q_j(t)}{1 + R_j^c(t+1)} \frac{\sum_{h=1}^N Q_h(t+1)P_h(t+1)}{\sum_{h=1}^N Q_h(t)P_h(t)} = \frac{Q_j(t)}{1 + R_j^c(t+1)} \sum_{h=1}^N w_{h,0,0} (1 + R_h(t+1)) \tag{A.17}$$

where the budget equality (A.3) with no net inflow (self-financing portfolio) has been used.

Inserting (A.16) into (A.10) gives:

$$1 + R_{CM}(T) \equiv \prod_{t=0}^{T-1} \sum_{j=1}^N w_{j,0,0} (1 + R_j(t+1)) \quad (A.18)$$

DISCUSSION QUESTIONS

1. Discuss the main reasons for rebalancing the assets in the financial portfolio.
2. Explain the effects of fixed and proportional trading costs on optimal asset allocation.
3. Illustrate the problem of optimal rebalancing with respect to a given benchmark portfolio.
4. Describe the most popular rules of portfolio rebalancing and discuss the relative merits.
5. Explain the functioning of two popular trading strategies, portfolio insurance and life-cycle investments, aimed at protecting the portfolio.

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Fig. 18.1 Trading, monitoring and rebalancing

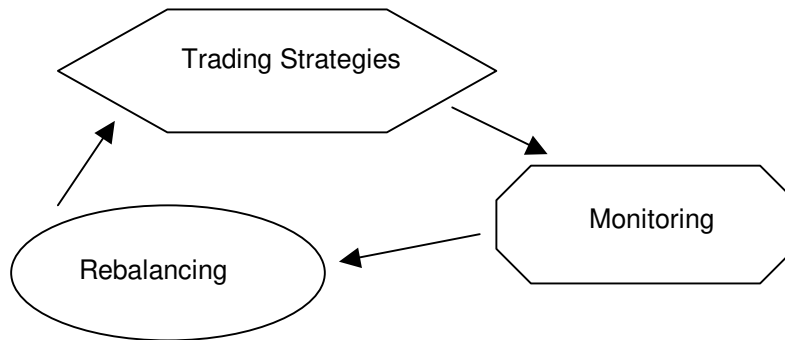


Fig. 18.2 Absolute and relative trading strategies

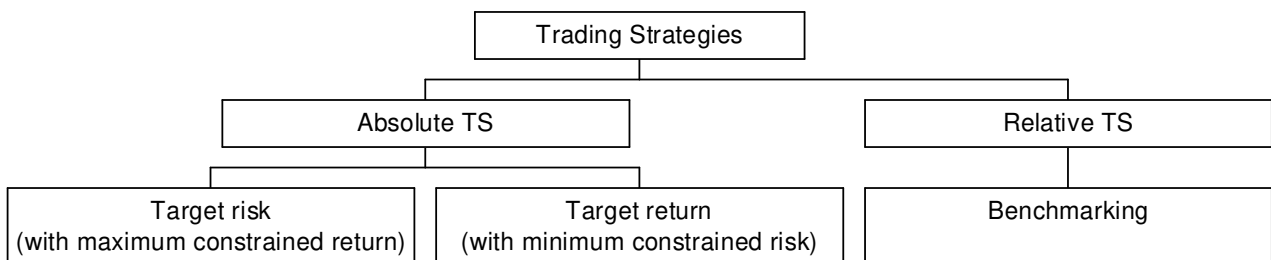


Fig. 18.3 Simulated values and changes of optimal portfolios with HARA utility

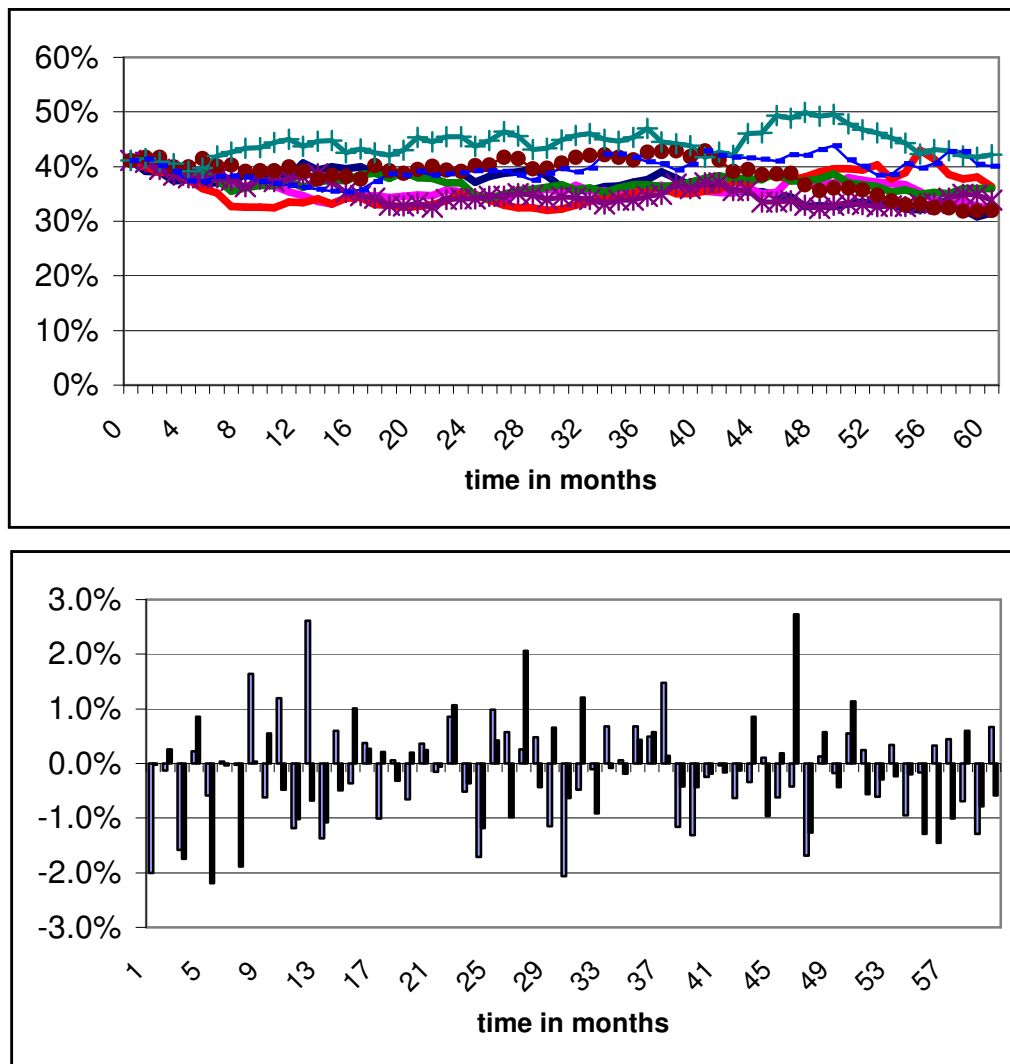


Fig. 18.4 No-trade wedge in the case of one risky asset

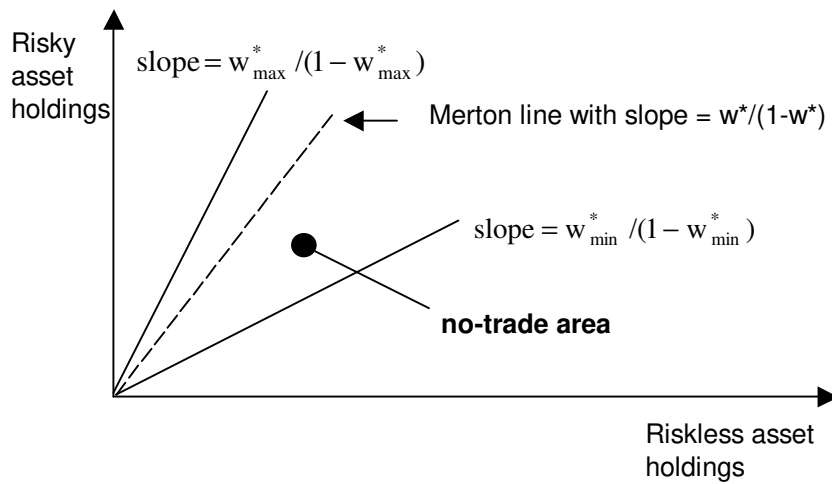


Fig. 18.5 Optimal allocation boundaries for the risky asset as function of transaction costs

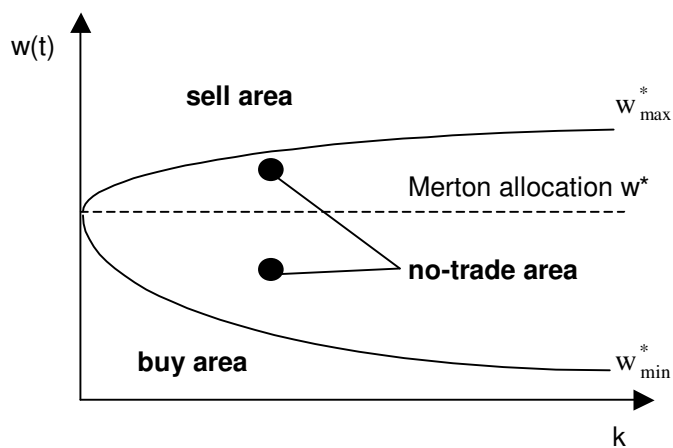


Fig. 18.6 No-trade area as function of proportional costs k for fixed costs $K > 0$

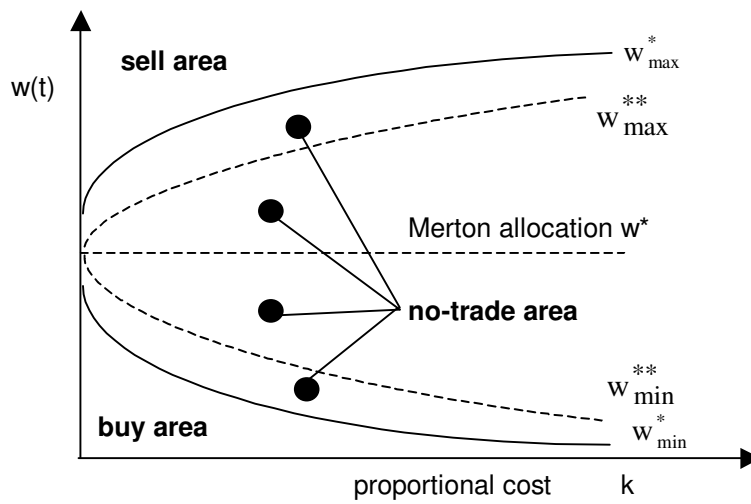


Fig. 18.7 Single-period portfolio rebalancing with transaction costs

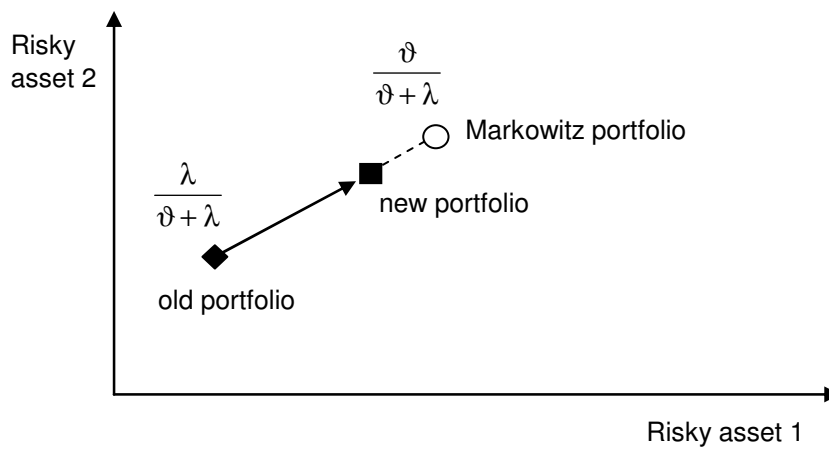


Fig. 18.8 Dynamic portfolio rebalancing with transaction costs

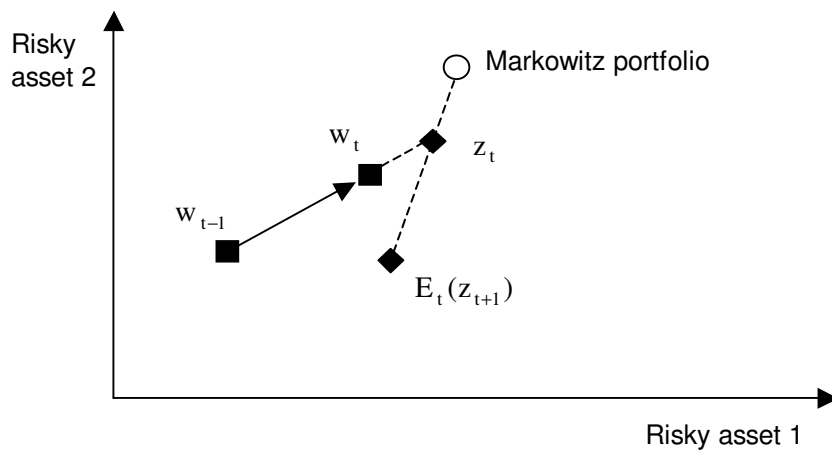


Fig. 18.9 Expected evolution of optimal portfolio with transaction costs and predictable returns

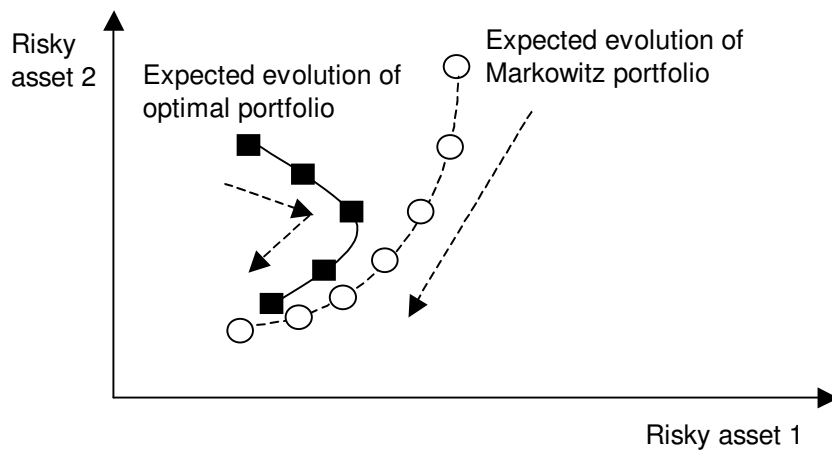


Fig. 18.10 No-trade regions with two risky assets

